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RESULTS

Results

01 · FRIEDMAN

nonparametric repeated measures

Table 1. Rank summary

Condition	Mean rank
score_t1	1.23
score_t2	2.04
score_t3	2.73

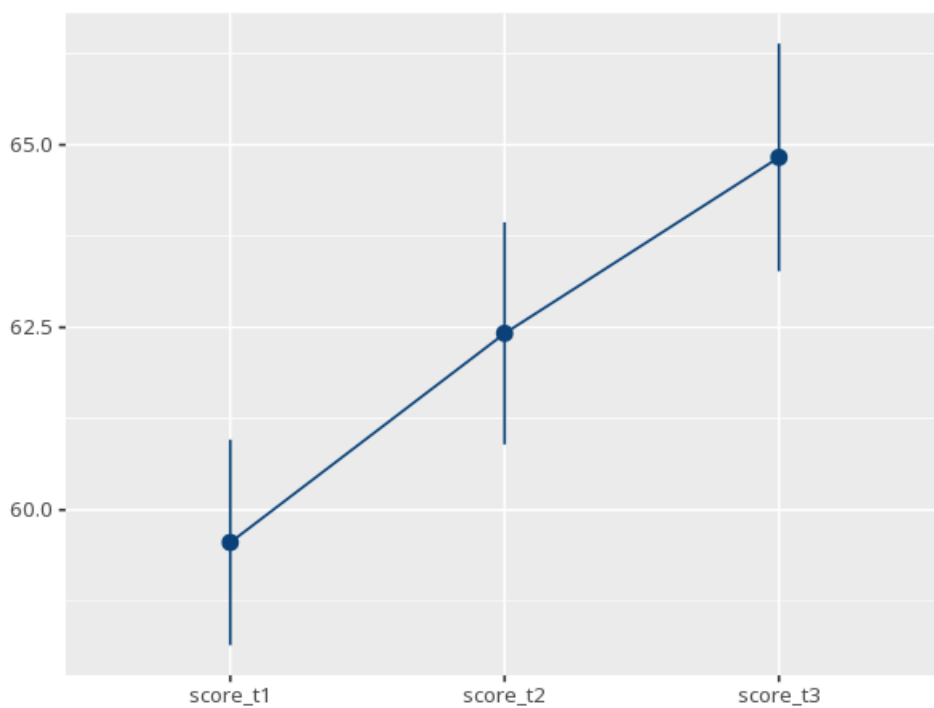
Table 2. Friedman test

χ^2	df	p	Kendall's W [95% CI]
67.19	2	<.001	0.56 [0.48, 1.00]

Table 3. Post-hoc (Nemenyi)

Pair	p_{adj}
score_t1 - score_t2	<.001
score_t1 - score_t3	<.001
score_t2 - score_t3	<.001

Figure. Across conditions



Understanding the numbers

χ^2 . The Friedman test statistic, testing whether ranks differ systematically across conditions. Here, $\chi^2(2) = 67.19$.

Condition. Which repeated-measures condition this row describes. Here, the conditions compared are score_t1, score_t2, score_t3.

df. The degrees of freedom for the Friedman test, based on the number of conditions. Here, $df = 2$.

Mean rank. The average rank of each condition's scores across subjects – the quantity the test compares. Here, the mean rank is highest for score_t3 (2.73) and lowest for score_t1 (1.23).

p. The probability of rank differences this large (or larger) across conditions if there were truly no systematic difference. Here, $p < .001$ - below the conventional .05 threshold, ranks differ across conditions.

p adj. The Nemenyi post-hoc adjusted p-value for a specific pair of conditions. Here, the strongest pairwise difference is score_t1 - score_t3, $p \text{ adj} < .001$.

Kendall's W. The effect size for the Friedman test – the degree of agreement in ranking across conditions, from 0 to 1. Here, $W = 0.56 [0.48, 1.00]$.

N. The number of subjects with complete data across every condition. Here, $N = 60$ subjects.

How to read this test

The nonparametric counterpart to repeated-measures ANOVA. The χ^2/p tests whether ranks differ across conditions; Kendall's W is the effect size (0–1 agreement). Post-hoc shows which conditions differ. Your significance threshold (α) is 0.05. $N = 60$ subjects.

APA template: A Friedman test gave $\chi^2(2)=67.19$, $p < .001$, $W=.56 [.48, 1.00]$.

Statistical basis: Friedman rank test (nonparametric counterpart to repeated-measures ANOVA) (Friedman, M. (1937). "The use of ranks to avoid the assumption of normality implicit in the analysis of variance." *Journal of the American Statistical Association*, 32(200), 675-701.); Kendall's W effect size (agreement among rankings) (Kendall, M. G. (1948). *Rank Correlation Methods*. Charles Griffin & Company.); Nemenyi post-hoc pairwise comparisons (Nemenyi, P. B. (1963). *Distribution-free multiple comparisons* [Doctoral dissertation, Princeton University].). Full references in the exported CITATIONS.txt.

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